Personal Finance Tracker – Project Report

1. Introduction

Managing personal finances effectively is essential for financial stability and long-term planning. This project aims to build a structured, database-driven Personal Finance Tracker that enables users to record income, expenses, track savings goals, and monitor monthly budgets using SQL.

The system is scalable, supports multiple users, and provides analytical insights into spending habits through SQL queries and views.

2. Abstract

This project involves designing a relational database in MySQL to manage and analyze personal finance data for multiple users. The system includes tables for Users, Categories, Income, Expenses, Goals, and Budget Adjustments. Dummy data for 10 users was inserted to simulate real-world usage. Key features include monthly summaries, category-wise analysis, goal tracking, and automated balance views. A Python script exports reports to CSV for further analysis.

3. Tools Used

Tool	Purpose
MySQL	Database engine
MySQL Workbench / CLI	Schema design, query execution
Python (with pandas, mysql.connector)	Report export
PDF Generator (optional)	Final report/documentation
ERD Generator	Entity Relationship Diagram

4. Steps Involved in Building the Project

- Database Schema Design
- Designed tables: Users, Categories, Income, Expenses, Goals, BudgetAdjustments.
- Established foreign key relationships for referential integrity.
- Dummy Data Insertion
- Created 10 sample users with various income/expense patterns.
- Inserted categorized income and essential/non-essential expenses.
- Defined personal financial goals for each user.
- SQL Queries and Views
- Generated category-wise and monthly summaries.
- Created views like MonthlyBalance to auto-calculate net savings.
- Added advanced queries for spending trends and goal progress.
- Exporting Reports
- Developed a Python script to connect with MySQL and export monthly reports in .csv format using pandas.

5. Conclusion

- The project successfully demonstrates how SQL can be used to create a comprehensive personal finance tracking solution. It empowers users to:

- Track expenses and income in real time
- Set and monitor savings goals
- Analyze spending behavior by category
- Export actionable insights via automated reports
- The modular design ensures scalability, while the use of standard SQL makes it easily portable to other relational databases.